Assignment No:1

Title – Design and implement of a Simple Reflex Agent for vaccum Cleaner World and Analysis of its Rationl Behaviour

Program –

```
environment = {
   'A': 'Dirty',
   'B': 'Dirty'
}
# Initial position location
= 'A'

# Path cost path_cost
= 0

def reflex_agent(location, status):

   if status == "Dirty":
return "Suck"     elif
location == "A":
      return "Right"
else:
      return "Left"

print("Initial Environment:", environment)
```

```
while True:
    status = environment[location]
action = reflex_agent(location, status)

    print(f"\nVacuum is at {location}")
print(f"Room Status : {status}")
print(f"Action Taken: {action}")

    if action == "Suck":
        environment[location] = "Clean"
path_cost += 1      # Cost for cleaning

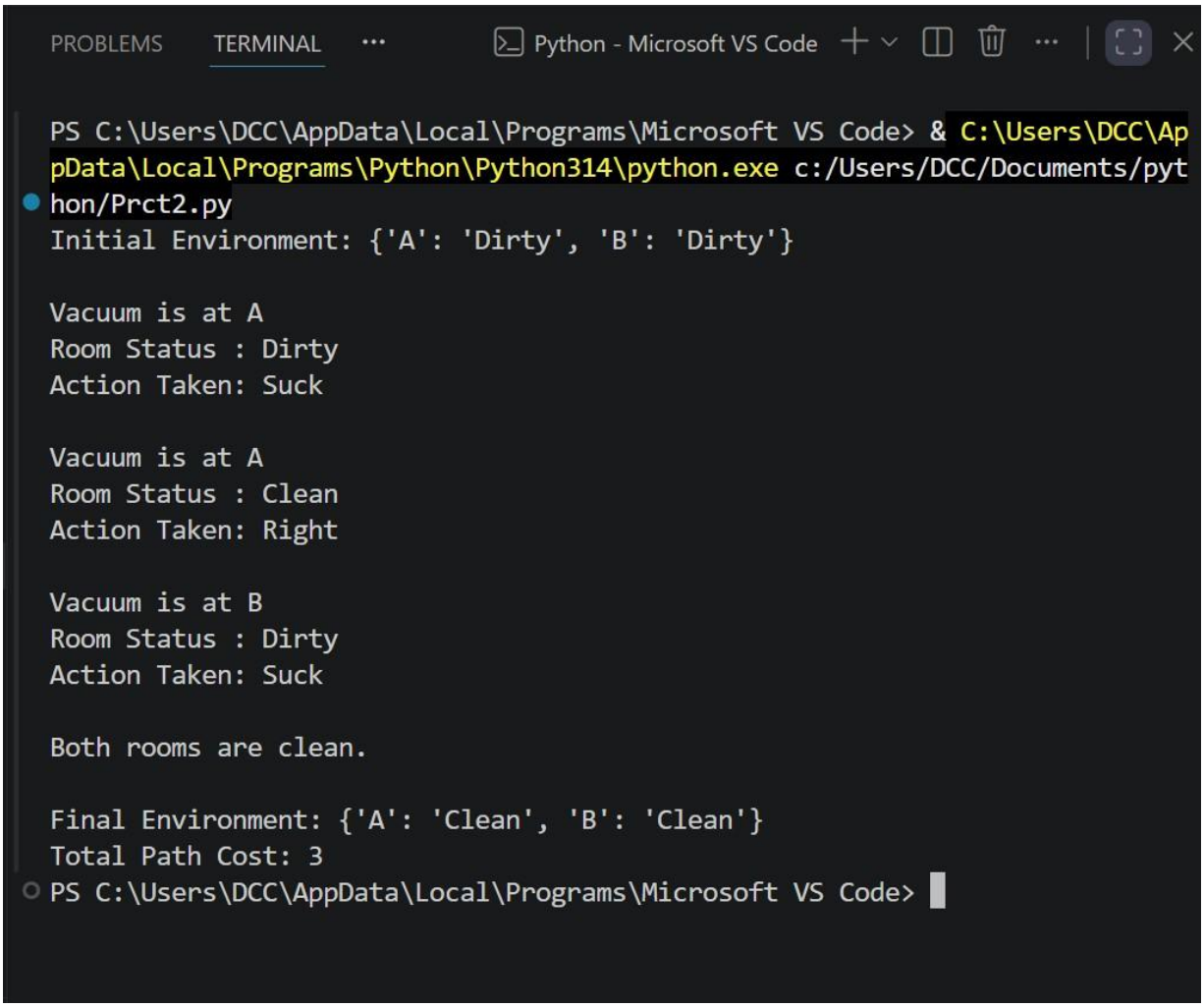
    elif action == "Right":
location = "B"         path_cost
+= 1

    elif action == "Left":
location = "A"         path_cost
+= 1

    if environment["A"] == "Clean" and environment["B"] == "Clean":
        print("\nBoth rooms are clean.")
break

print("\nFinal Environment:", environment)
print("Total Path Cost:", path_cost)
```

Output –


PROBLEMS
TERMINAL
Python - Microsoft VS Code
PS C:\Users\DCC\AppData\Local\Programs\Microsoft VS Code> & C:\Users\DCC\AppData\Local\Programs\Python\Python314\python.exe c:/Users/DCC/Documents/python/Prct2.py
Initial Environment: {'A': 'Dirty', 'B': 'Dirty'}
Vacuum is at A
Room Status : Dirty
Action Taken: Suck
Vacuum is at A
Room Status : Clean
Action Taken: Right
Vacuum is at B
Room Status : Dirty
Action Taken: Suck
Both rooms are clean.
Final Environment: {'A': 'Clean', 'B': 'Clean'}
Total Path Cost: 3
PS C:\Users\DCC\AppData\Local\Programs\Microsoft VS Code>